

The mass estimator equation

A positive inertial-mass estimate from measured force and acceleration, with uncertainty in both vectors.

THE COMPACT EQUATION / GAUSSIAN ERRORS + FLAT POSITIVE MAGNITUDES

$$\hat{m} = \frac{E_P[f]}{E_P[\alpha]} = \frac{\int_C f e^{-Q/2} dv}{\int_C \alpha e^{-Q/2} dv} \quad \text{Average the magnitudes, then divide.}$$

$$C = (0, \infty)^2 \times S^{d-1} \quad dv = df d\alpha d\Omega(u) \quad d = 1, 2 \text{ or } 3$$

$$dP = Z^{-1} e^{-Q/2} dv \quad Z = \int_C e^{-Q/2} dv$$

$$Q = (y - x)^T \Sigma^{-1} (y - x) \quad \Sigma: \text{symmetric positive-definite covariance}$$

$$y = \begin{bmatrix} \tilde{\mathbf{F}} \\ \tilde{\mathbf{a}} \end{bmatrix} \quad x = \begin{bmatrix} fu \\ \alpha u \end{bmatrix} \quad f > 0, \alpha > 0, ||u|| = 1$$

Uniform direction; flat in f and α .

P is the normalized law over compatible latent pairs. F and a above are measured vectors.

Types and form

The inputs are two **vectors** and their joint covariance **matrix**. The output is a positive **scalar mass**, accompanied by a conditional interval and a logarithmic spread. The code supports one, two or three spatial dimensions.

The integration variables f and α are possible true magnitudes; u is their common unit direction. The formula averages these magnitudes under one joint probability law, then divides. Its mathematical form is a ratio of weighted integrals.

Basic physical assumption

Newton II holds for the true quantities: $F^* = m a^*$, with $m > 0$. Both measurements refer to the same object, time or interval, and inertial frame. Neither input may already assume the mass being sought.

Statistical assumptions that complete the equation

Measurement errors are jointly Gaussian with zero mean and the supplied positive-definite covariance. The common direction is uniform, and the reference measure is flat in the two positive physical magnitudes. These choices are explicit parts of the estimator; Newton II alone does not determine them.

Inputs and uncertainty

Preserve the vector components and specify their errors in the same coordinate system.

Input	Meaning and units
force shape (d,)	Measured net force components. SI unit: N. Zero and negative components are valid.
acceleration shape (d,)	Measured acceleration components in the same axes. SI unit: m/s ² .
covariance shape (2d, 2d)	Error covariance in [F1, ..., Fd, a1, ..., ad] order. Diagonal entries are variances. Supply cross-correlations when present.
probability default 0.95	Probability content of the central equal-tail interval. Must lie strictly between zero and one.

For independent isotropic 3D errors, use diagonal entries $[\sigma_F^2, \sigma_F^2, \sigma_F^2, \sigma_a^2, \sigma_a^2, \sigma_a^2]$. Each σ is a **per-coordinate standard uncertainty**. The covariance block units are N², (m/s²)² and N·m/s².

Uncertainty comes from the same joint law

Let $M = f/\alpha$ under P , and let q_p be its probability- p quantile. The returned uncertainty is:

95% conditional interval: $[q_{0.025}, q_{0.975}]$
Logarithmic spread: $SD_p[\log(M/m_0)]$

The reference mass m_0 makes the logarithm dimensionless; its choice does not change the spread. A 95% interval contains 95% of this declared conditional law. It does **not automatically promise 95% coverage** across repeated experiments.

The ordinary mean and second moment of M diverge in this model, so an ordinary mass standard deviation is undefined. Quantiles and log spread remain finite. Report the interval as endpoints, not as a symmetric mass $\pm$ SD.

Use and interpretation

1. Establish the object, frame, interval, units and covariance.
2. Supply the measured vector components directly.
3. Report the point, interval, probability and declared model together.
4. Keep the original measurements and covariance with their provenance.

The point need not equal the median or interval midpoint. Inverse mass uses $1/\hat{m}$ and the reversed reciprocal interval $[1/\text{upper}, 1/\text{lower}]$.

Calculate and report it

A standalone Python function is included in this folder. It requires Python 3.10+ and NumPy.

Install: `python -m pip install -r requirements.txt`

Run the supplied example: `python example.py`

```
import numpy as np
from mass_estimator import estimate_mass

force = [10.0, 0.0, 0.0]          # N
acceleration = [2.0, 0.1, 0.0] # m/s^2
cov = np.diag([0.5**2]*3 + [0.1**2]*3)
r = estimate_mass(force, acceleration, cov)
print(r.mass, r.interval)
print(r.log_standard_deviation)
```

Output	Example value
Mass point	4.993754 kg
95% conditional interval	[4.344146, 5.740363] kg
Log standard deviation	0.071035 (dimensionless)

Scope and numerical checks

The function accepts full covariance, including anisotropy and correlated channels. An optional externally known direction replaces the direction average. The supplied covariance is treated as fixed; uncertainty in its calibration requires the separate extended model.

Measured zeros and nonaligned vectors are valid inputs. Exact quantities, hard bounds and changing-excitation joint trials require separate treatment. At true zero excitation, measurements cannot identify mass; a finite model-based output does not establish identification.

Every returned result passes a grid-refinement comparison. The example's final scaled change is 6.61×10^{-5} ; this is a numerical diagnostic, not mass uncertainty or a certified error bound. Unresolved integration raises an error. Nine verification checks passed, including analytic identities and independent integrations.

Which version this specifies

The current **ratio of magnitude means** is retained in the local 21 September synthesis. The earlier best-point and median proposals were checked. The 7 September steering assumes Newton II and removes empirical rejection from this conditional use.

Managed sources: [working specification §§7-12, 22-23](#); [7 September steering](#); [readout audit](#). VD-docs commit: 8782a293297330f7a354d87ebe62792018fa8866. Full source paths, local source hashes, version distinctions and the API contract are in README.md.

Status: user-requested presentation, staged in pass to VD-docs; awaiting transfer.